

$$1. 76_{10} \Rightarrow 1001100_2 \Rightarrow 114_8 \Rightarrow 4C$$

$$\begin{array}{cccccc} 0 & 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 2 & 4 & 8 & 16 & 32 & 64 \\ 4 & + & 8 & + & 64 & = & 76_{10} \end{array}$$

$$\begin{array}{ccc} 4 & 1 & 1 \\ 1 & 8 & 64 \\ \downarrow & \downarrow & \downarrow \\ 4 & + & 8 & + & 64 & = & 76_{10} \end{array}$$

$$\begin{array}{cc} C & 4 \\ 1 & 16 \\ \downarrow & \downarrow \\ 12 & + & 64 & = & 76_{10} \end{array}$$

$$2. 11000100_2 \Rightarrow 196_{10} \Rightarrow C4_{16} \Rightarrow 304_8$$

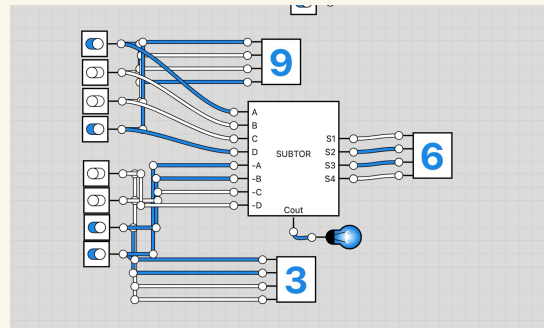
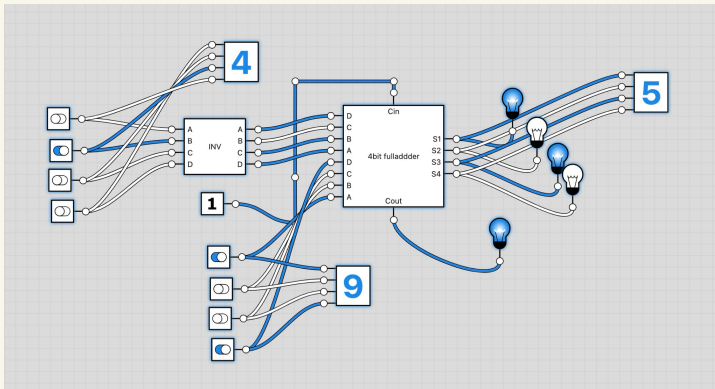
$$128 + 64 + 32 + 16 + 8 + 4 + 2 + 1$$

$$128 + 64 + 4 = 196$$

$$(C \times 16) + (4 \times 1) = 196$$

$$\begin{array}{l} 196_{10} \div 8 \text{ เศษ } 4 \\ 24 \div 8 \text{ เศษ } 0 \\ 3 \div 8 \text{ เศษ } 3 \end{array}$$

$$304_8$$

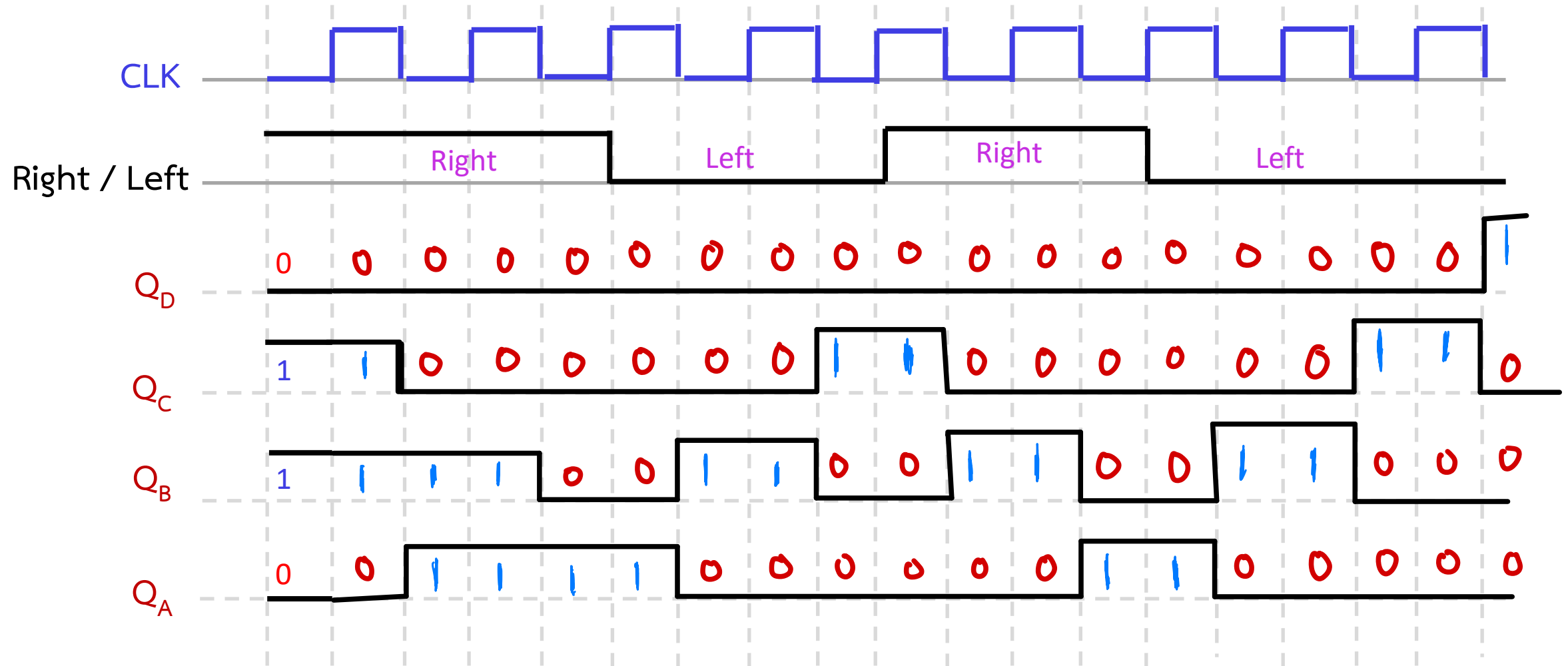


เอาเลขที่ต่อการลบ กลับ บิตและ + 1 และรวมเข้า

Full Adder ในตัวอย่าง 9-5 = 4

$$1001_2 + 0100_2 + 1 = 0101_2$$

Quiz: Please complete the timing diagram of 4-bits Bidirectional Shift Register



$$\begin{aligned}
 4. \quad A\bar{B} + CA + C\bar{D} &= \overline{\overline{A\bar{B} + CA + C\bar{D}}} \\
 &= (\overline{A\bar{B}} \cdot \overline{CA} \cdot \overline{C\bar{D}}) \\
 &= (\bar{A} + B)(\bar{C} + \bar{A})(\bar{C} + D)
 \end{aligned}$$

$$5 \quad \Sigma(0, 5, 6, 10, 12, 15)$$

Don't care

$$\Sigma(2, 4, 7, 8, 14)$$

Sum of Product

AB \ CD	00	01	11	10
00	1	X	X	0
01	0	1	X	1
11	1	0	1	X
10	X	0	0	1

$\bar{A}\bar{D} + \bar{B}\bar{C}\bar{D} + BC + \bar{A}D$

6.

6.) จงออกแบบวงจร Synchronous Counter Circuit โดยใช้ NOR Gate RS Flip-Flop ในการนับตัวเลขเรียงลำดับแบบวนซ้ำ ตามลำดับ 0 → 1 → 2 → 3 → 4 → 0 โดยแสดงขั้นตอนการได้มาซึ่งวงจรผลลัพธ์ให้ชัดเจน (25 points)

R	S	Q_{n+1}
0	0	1
0	1	0
1	0	0
1	1	1

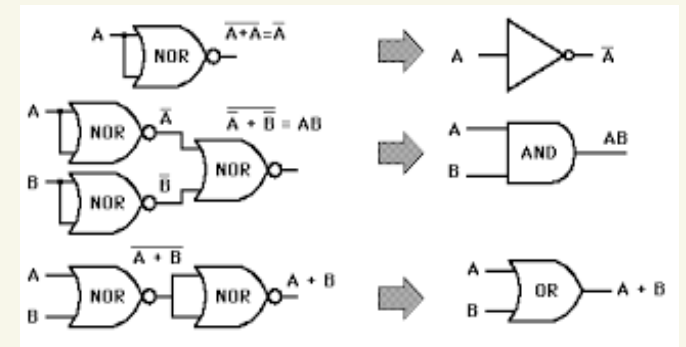
Q_n	Q_{n+1}	R	S
0	0	x	0
0	1	0	1
1	0	1	0
1	1	0	x

Q_n	Q_{n+1}
0 0 0	0 0 1
0 0 1	0 1 0
0 1 0	1 0 0
1 0 0	0 0 0

R_A	S_A	R_B	S_B	R_C	S_C
x	0	x	0	0	1
x	0	0	1	1	0
0	1	1	0	x	0
1	0	x	0	x	0

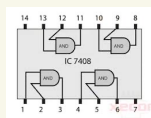
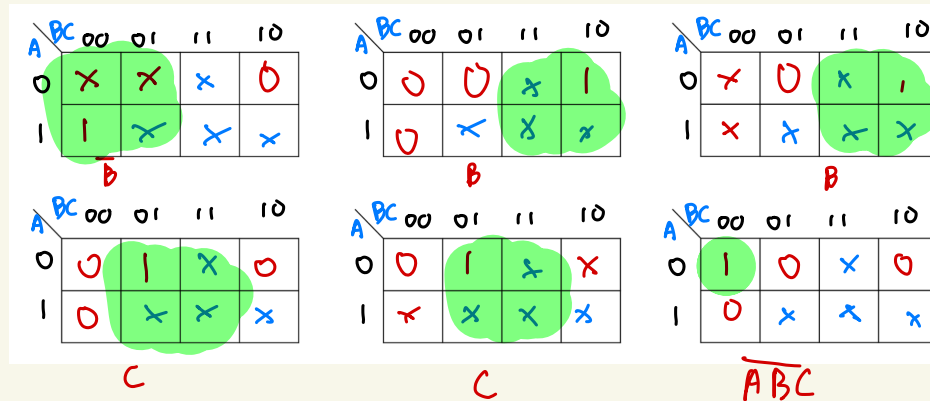
NOR → ∞

คู่มือข้อ 4 โฉม 1



NAND → ∞

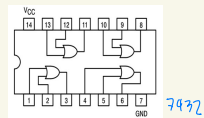
คู่มือข้อ 4 โฉม 1



AND

A	B
00	0
01	0
10	0
11	1

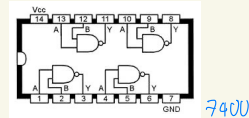
$A \cdot B$



OR

A	B
00	0
01	1
10	1
11	1

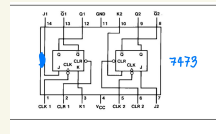
$A + B$



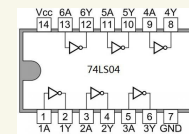
NAND

A	B
00	1
01	1
10	1
11	0

$\overline{AB}$



CLK ลง ↓



NOT

Input	Output
0	1
1	0

